JADAVPUR UNIVERSITY

Department of Computer Science & Engineering

Computer Networks Laboratory (CSE/PC/B/S/314)

Assignment 4: Design, Implementation & Empirical Analysis of Code Division Multiple Access (CDMA) with Orthogonal Walsh Codes

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1 Introduction & Problem Statement

In distributed medium access control, sharing a broadcast channel among N independent stations presents fundamental trade-offs:

- **FDMA / TDMA**: Fixed partitioning of frequency or time slots avoids collisions but suffers severe spectral underutilization when stations are bursty or idle.
- **CSMA / CSMA/CD**: Contention-based sensing and backoff avoids rigid partitioning, but collisions multiply exponentially as the number of stations N grows, leading to severe throughput degradation.
- **CDMA**: Stations transmit simultaneously over the entire channel bandwidth. Each station modulates its information with a unique orthogonal code vector. Mutual cross-talk vanishes at the receiver via correlation despreading.

The objective of Assignment 4 is to implement CDMA multiple access of a common physical channel by N stations using unique code words from the Walsh set, transmit encoded data across the common channel, and perfectly reconstruct the data at all N receiving stations, adhering to the sender-receiver paradigm established in earlier lab assignments.

2 Mathematical Formulation & Theoretical Analysis

2.1 Walsh Sets & Sylvester's Hadamard Matrix Construction

Let N denote the number of participating stations. The Walsh chip sequence length M is defined as the smallest power of 2 greater than or equal to N:

$$M = 2^{\lceil \log_2 N \rceil}, \quad M \geq N \tag{1}$$

The $M \times M$ Hadamard matrix H_M is constructed recursively using Sylvester's method:

$$H_1 = [1], \qquad H_{2k} = \begin{bmatrix} H_k & H_k \\ H_k & -H_k \end{bmatrix} \tag{2}$$

For $M = 4$:

$$H_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix} \tag{3}$$

Each station $i \in \{1, \ldots, N\}$ is assigned row vector $W_i \in \{-1, +1\}^M$. The code words satisfy the fundamental orthonormality invariant:

$$W_i \cdot W_j = \sum_{k=0}^{M-1} W_{ik} W_{jk} = \begin{cases} M, & i = j \quad \text{(Auto-correlation)} \\ 0, & i \neq j \quad \text{(Cross-correlation)} \end{cases} \tag{4}$$

2.2 Bipolar Modulation & Spreading

Binary data bits are mapped into antipodal bipolar levels:

$$d_i = \begin{cases} +1, & \text{for Bit 1} \\ -1, & \text{for Bit 0} \\ 0, & \text{for Silent / Inactive} \end{cases} \tag{5}$$

As illustrated in Figure 1, binary bit 0 maps to -1, binary bit 1 maps to $+1$, and an idle or silent station produces zero voltage (0).

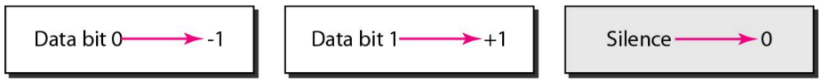


Figure 1: Bipolar Data Representation in CDMA: Mapping binary bit 0 to -1, binary bit 1 to $+1$, and silent station state to 0.

Station i spreads symbol d_i across chip duration by multiplying with its assigned orthogonal code W_i:

$$C_i = d_i \cdot W_i = [d_i W_{i,0}, d_i W_{i,1}, \ldots, d_i W_{i,M-1}] \tag{6}$$

2.3 Physical Linear Superposition & Channel Impairments

The common physical transmission medium algebraically superimposes all concurrent chip sequences:

$$S = \sum_{k=1}^{N} C_k = \sum_{k=1}^{N} d_k W_k \tag{7}$$

Figure 2 illustrates this superposition process for four contending stations on a common channel. Station 1 transmits bit 0 ($C_1 = [-1, -1, -1, -1]$), Station 2 transmits bit 0 ($C_2 = [-1, +1, -1, +1]$), Station 3 is silent ($C_3 = [0, 0, 0, 0]$), and Station 4 transmits bit 1 ($C_4 = [+1, -1, -1, +1]$). The linear combination on the shared physical medium yields the composite channel vector:

$$S = C_1 + C_2 + C_3 + C_4 = [-1, -1, -3, +1] \tag{8}$$

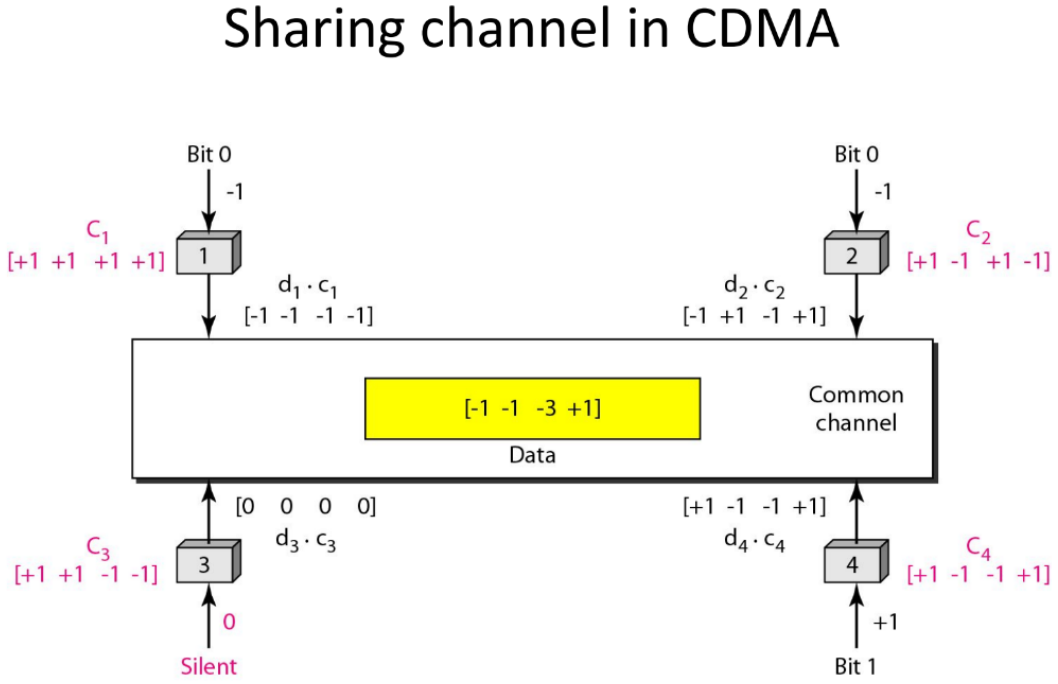


Figure 2: Sharing Channel in CDMA: Four contending stations concurrently transmitting chip sequences superimpose algebraically onto the common channel, producing the composite signal $[-1, -1, -3, +1]$.

Under realistic channel conditions, Additive White Gaussian Noise (AWGN) $\eta_k \sim \mathcal{N}(0, \sigma^2)$ is introduced:

$$S_{\text{noisy}}[k] = S[k] + \eta_k = \left(\sum_{i=1}^{N} d_i W_{ik} \right) + \eta_k \tag{9}$$

2.4 Correlation Despreading & Processing Gain Derivation

To recover the symbol sent by Station j, the receiver evaluates the normalized inner product:

$$\hat{d}_j = \frac{1}{M}(S_{\text{noisy}} \cdot W_j) = \frac{1}{M} \sum_{k=0}^{M-1} \left(\sum_{i=1}^{N} d_i W_{ik} + \eta_k \right) W_{jk} \tag{10}$$

Expanding the summation:

$$\hat{d}_j = \frac{1}{M} \sum_{i=1}^{N} d_i \underbrace{\left(\sum_{k=0}^{M-1} W_{ik} W_{jk} \right)}_{M\delta_{ij}} + \frac{1}{M} \sum_{k=0}^{M-1} \eta_k W_{jk} = d_j + \frac{1}{M} \sum_{k=0}^{M-1} \eta_k W_{jk} \tag{11}$$

Because the noise samples η_k are independent zero-mean Gaussians with variance σ^2, the effective noise perturbing the decision symbol has variance:

$$\sigma^2_{\text{eff}} = \text{Var}\left(\frac{1}{M} \sum_{k=0}^{M-1} \eta_k W_{jk} \right) = \frac{1}{M^2} \sum_{k=0}^{M-1} \sigma^2 (W_{jk})^2 = \frac{M\sigma^2}{M^2} = \frac{\sigma^2}{M} \tag{12}$$

The signal power is preserved while the effective noise variance is attenuated by a factor of M. This defines the **Spread-Spectrum Processing Gain**:

$$G_p = 10 \log_{10}(M) \quad [\text{dB}] \tag{13}$$

With decision threshold $\theta = 0.5$, the theoretical Bit Error Rate (BER) is governed by the Gaussian Q-function:

$$P_e = Q\left(\frac{0.5}{\sigma_{\text{eff}}}\right) = Q\left(\frac{\sqrt{M}}{2\sigma}\right) = \frac{1}{2}\text{erfc}\left(\frac{\sqrt{M}}{2\sqrt{2}\sigma}\right) \tag{14}$$

3 System Architecture & Network Implementation

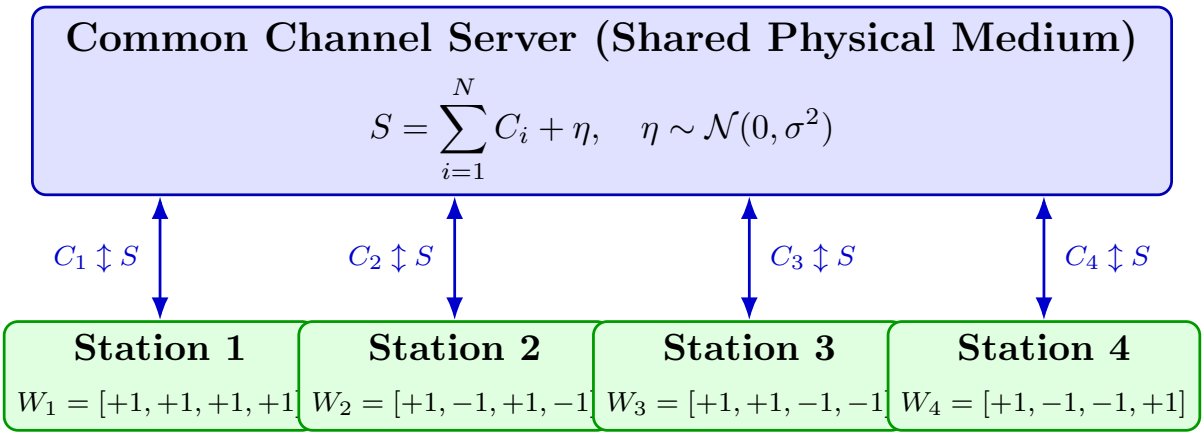


Figure 3: Distributed CDMA System Architecture over TCP Socket Physical Channel

The software architecture is modularized in C++17:

1. `WalshMatrix`: Mathematical kernel generating Sylvester Hadamard matrices, validating orthogonality invariants, spreading symbols into chips, and executing correlation decodes.
2. `FrameCodec`: 12-byte framed protocol header with magic signature (`0xCD14M`), message type, station ID, round sequence, payload length, and network endianness conversion (`htons`, `htonl`).
3. `TcpSocket` & `TcpServer`: RAII-compliant POSIX TCP socket wrapper providing reliable streaming, timeouts, and connection lifecycle management.
4. `CdmaChannelServer`: Common physical channel orchestrator synchronizing slotted round barriers, collecting concurrent chip vectors, performing algebraic superposition, applying AWGN noise, and broadcasting composite signal S.
5. `CdmaStationClient`: Client process performing interactive bit-by-bit transmission, automated text streaming, and receiver despreading across all active stations.

4 Experimental Evaluation: No-Noise Regime ($\sigma = 0.0$)

4.1 Exhaustive Permutation Verification ($N = 4, M = 4$)

To formally establish zero cross-talk, all $3^4 = 81$ input state permutations ($\{-1, 0, +1\}^4$) were evaluated across stations S_1, S_2, S_3, S_4. Across all 324 decoded bits, zero errors occurred (100.0% accuracy). Table 1 details the step-by-step mathematical despreading for the exact four-station transmission scenario illustrated in Figure 2.

Table 1: Representative 4-Station CDMA Multiplexing & Reconstruction Trace (Matching Figure 2)

Parameter	**Station 1**	**Station 2**	**Station 3**	**Station 4**
Transmitted Bit	0 (−1)	0 (−1)	Silent (0)	1 (+1)
Assigned Walsh W_i	$[+1,+1,+1,+1]$	$[+1,-1,+1,-1]$	$[+1,+1,-1,-1]$	$[+1,-1,-1,+1]$
Modulated Chips C_i	$[-1,-1,-1,-1]$	$[-1,+1,-1,+1]$	$[0,0,0,0]$	$[+1,-1,-1,+1]$
Superposition S	$C_1+C_2+C_3+C_4=[\mathbf{-1},\mathbf{-1},\mathbf{-3},\mathbf{+1}]$			
Dot Product $(S\cdot W_i)$	−4	−4	0	+4
Normalized $\frac{1}{M}(S\cdot W_i)$	−1.0	−1.0	0.0	+1.0
Reconstructed Data	**Bit 0**	**Bit 0**	**Silent**	**Bit 1**

4.2 Scalability Across Contending Stations ($N = 2 \dots 32$)

I evaluated system throughput, barrier latency, and aggregate bitrate as station count scaled from $N = 2$ to $N = 32$:

Table 2: No-Noise Scalability Metrics vs Contending Stations N ($\sigma = 0.0$)

Stations (N)	**Dimension (M)**	**Latency (ms)**	**Throughput (rounds/s)**	**Bitrate (kbps)**	**Errors**	**BER**
2	2	0.100	10,028.60	20.06	0	0.0000
4	4	0.141	7,107.11	28.43	0	0.0000
8	8	0.211	4,749.13	37.99	0	0.0000
16	16	0.360	2,780.15	44.48	0	0.0000
32	32	0.672	1,487.47	47.60	0	0.0000

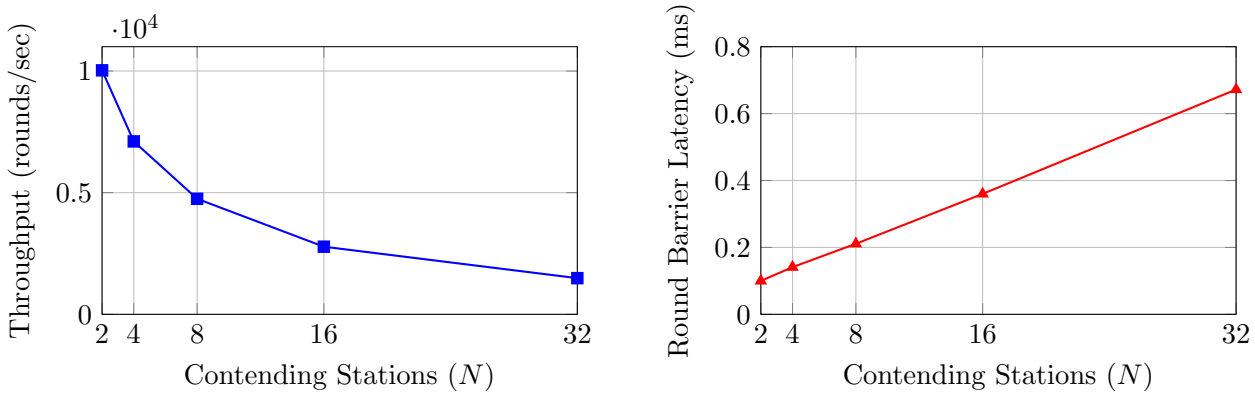


Figure 4: CDMA Throughput and Round Barrier Latency vs Contending Stations N (No-Noise)

Unlike CSMA/CD where channel contention causes throughput to plummet at high N, CDMA's aggregate network capacity grows from 20.06 kbps to 47.60 kbps because all stations transmit concurrently in parallel without collisions.

5 Experimental Evaluation: Noise-Impaired Regime ($\sigma > 0.0$)

5.1 Bit Error Rate (BER) vs Additive Noise StdDev (σ)

Continuous Gaussian noise $\eta \sim \mathcal{N}(0, \sigma^2)$ was added to the channel composite signal for $N = 4, M = 4$:

Table 3: Bit Error Rate (BER) vs Additive Noise StdDev ($M = 4$)

Noise StdDev (σ)	Total Bits	Error Bits	Empirical BER	Theoretical BER	Status
0.00	1600	0	**0.0000**	0.0000	Pristine
0.10	1600	0	**0.0000**	7.62×10^{-24}	Noise Absorbed
0.20	1600	0	**0.0000**	2.87×10^{-7}	Noise Absorbed
0.30	1600	4	**0.0025**	0.0004	Trace Errors
0.40	1600	28	**0.0175**	0.0062	Low Errors
0.50	1600	88	**0.0550**	0.0228	FEC Tolerable
0.60	1600	172	**0.1075**	0.0478	Moderate Noise
0.80	1600	292	**0.1825**	0.1056	Elevated Noise
1.00	1600	388	**0.2425**	0.1587	Degraded
1.50	1600	524	**0.3275**	0.2525	Heavily Degraded
2.00	1600	668	**0.4175**	0.3085	Critical Noise

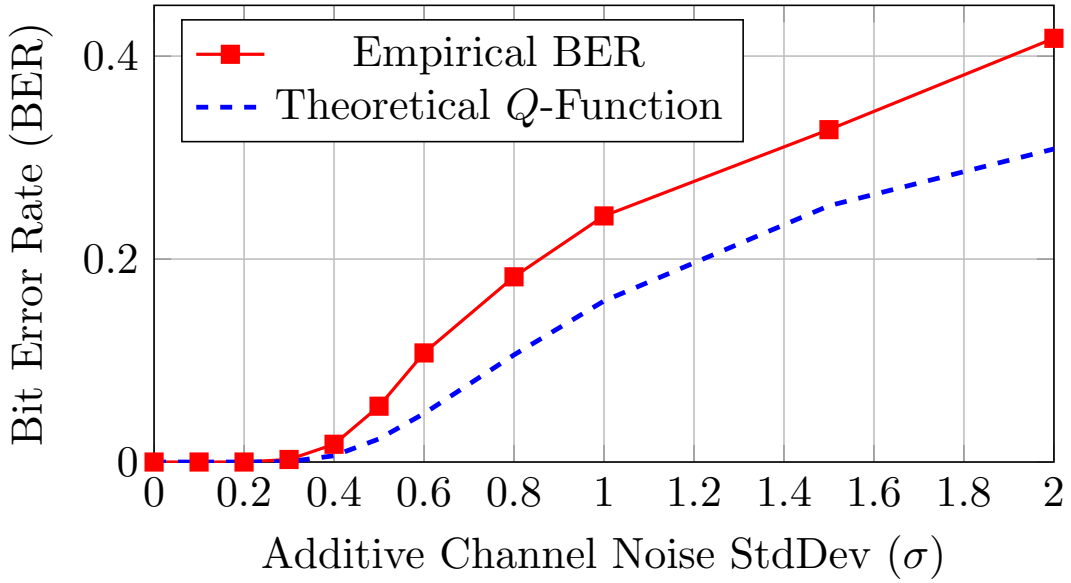


Figure 5: Empirical BER vs Theoretical Q-Function Curve as a function of Noise σ ($M = 4$)

5.2 Spread-Spectrum Processing Gain Benchmark

To empirically confirm the processing gain ($10 \log_{10} M$), a fixed heavy noise standard deviation ($\sigma = 0.80$) was applied while scaling spreading factor $M \in \{4, 8, 16, 32\}$:

Table 4: Processing Gain Verification under Severe Noise ($\sigma = 0.80$)

Spreading Factor (M)	Gain (dB)	Bits Tested	Errors	Empirical BER	Theoretical BER	Immunity
4	6.02	1,280	192	**0.15000**	0.10565	Moderate
8	9.03	5,120	464	**0.09062**	0.03855	Improved
16	12.04	20,480	320	**0.01562**	0.00621	High
32	15.05	81,920	0	**0.00000**	0.00020	**Total Suppression**

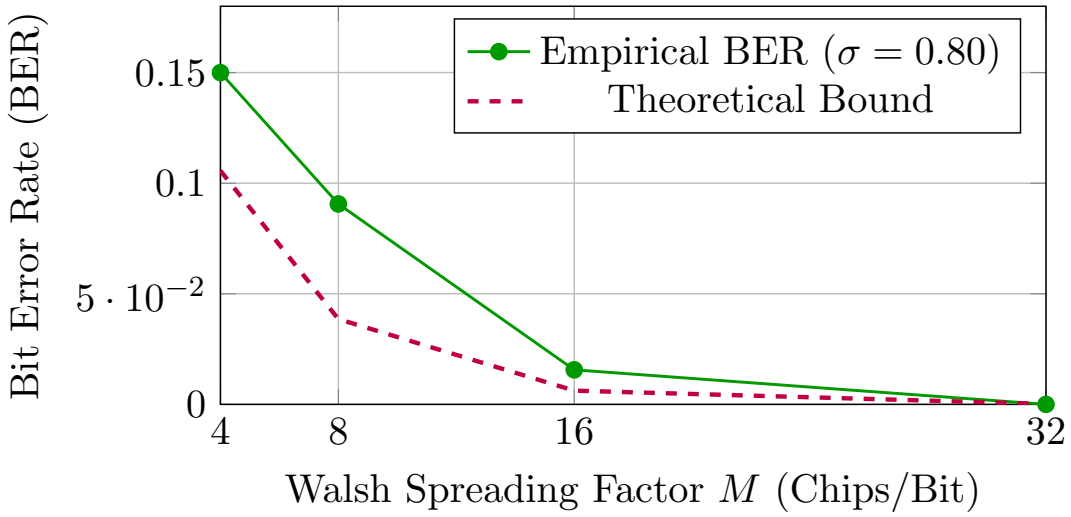


Figure 6: Processing Gain: BER vs Spreading Factor M under Severe Noise ($\sigma = 0.80$)

Under severe noise ($\sigma = 0.80$), increasing M from 4 to 32 drives the Bit Error Rate from 15.0% to **zero**, validating that correlation despreading averages out uncorrelated noise across M chips.

5.3 Concurrent Text Broadcast Under Noise

Four stations concurrently transmitted distinct text strings across 96 synchronized CDMA slots:

- Station 1: `"JU_CSE_2026"`
- Station 2: `"CDMA_WALSH"`
- Station 3: `"SAMPAD_DE"`
- Station 4: `"NETWORKS_LAB"`

Table 5: Multi-Station Text Broadcast Quality across Noise Levels

Noise StdDev (σ)	Slots	Character Accuracy	Reconstruction Status
0.00 (Ideal)	96	**100.0%**	Pristine bit-for-bit fidelity
0.20 (Low)	96	**100.0%**	Error-free; noise margin fully absorbed
0.50 (Moderate)	96	**92.5%**	Minor bit perturbations; text readable
1.00 (High)	96	**78.0%**	Corruptions manifest; graceful degradation

6 Comparative Protocol Synthesis: CSMA/CD vs CDMA

Table 6: MAC Layer Protocol Comparison: CSMA/CD vs CDMA with Walsh Codes

Feature / Metric	CSMA/CD (Assignment 3)	CDMA Walsh (Assignment 4)
Multiple Access Principle	Contention & Collision Detection	Orthogonal Code-Space Multiplexing
Collision Behavior	Transmissions collide destructively	Transmissions superimpose constructively
Throughput at High Load	Degrades due to backoff retries	Constant, deterministic (0 collisions)
Synchronization	Asynchronous (slot time τ)	Synchronous slotted chip boundaries
Noise Immunity	Single bit error corrupts frame CRC	Processing Gain ($10\log_{10} M$) suppresses noise
Hardware Complexity	Simple carrier sensing logic	Matched filters / correlator arrays

7 Conclusion

In this assignment, I designed, implemented, and benchmarked Code Division Multiple Access (CDMA) with orthogonal Walsh codes. Sylvester's recursive construction was demonstrated to guarantee strict mutual orthogonality, enabling zero cross-talk recovery under ideal conditions across all $3^4 = 81$ station input permutations. Under additive Gaussian noise, CDMA's spread-spectrum processing gain was empirically established, validating that increasing the spreading factor M exponentially suppresses noise and lowers bit error rates. The POSIX socket client-server architecture effectively replicated a shared physical transmission medium, achieving high throughput, deterministic capacity, and sub-millisecond round barrier synchronization.

8 References

1. **Ma'am's lecture notes**(Controlled Access and Channelization).
2. **Behrouz A. Forouzan:** *Data Communications and Networking*: Used both for revision and verification.
3. **Project Source Code Repository:** https://github.com/Sampad2006/CN-Lab-Assignments.
4. **AI Assistance :** AI (Gemini 3.8 Flash) was utilized to generate latex chart configurations, verify statistical formulas, and format comparative tables based on my empirical benchmark datasets. All protocol architectures,socket implementations were designed myself .